Laura S. Bumbulis

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ORCiD

Education, University of Waterloo

PhD Statistics 2024–present
Thesis: *Design and Analysis of Studies Involving Incompletely Observed Multistate Processes*
Advisor: Richard J. Cook
Overall Average (6 courses): 96%

MMath Statistics 2023–2024
Masters Research Paper: *Robustness and Efficiency Considerations when Testing Process Reliability with a Limit of Detection*
Advisor: Richard J. Cook
Overall Average (7 courses): 97%

BMath (Co-op) 2017–2022
Majors: Statistics; Combinatorics & Optimization
Minor: Pure Mathematics
Overall Average (45 courses): 94%

Publications

5. Eder, L., Shahab, S., Gillespie, S., **Bumbulis, L.**, Emanoilidis, H., Compher, C., Scher, J., Gladman, D., Cook, R., Chandran, V., & Ogdie, A. (2026). Dietary interventions in psoriatic arthritis: a randomized controlled clinical trial. *The Journal of Rheumatology*, *53* (Suppl 1), 138. DOI:10.3899/jrheum.2026-0447.177
4. Cook, R. J., **Bumbulis, L. S.**, Kim, S., & Albert, P. S. (2026). Modeling the effects of smoking on lung cancer. *Observational Studies*, *12* (2), 193–209. muse.jhu.edu/pub/56/article/999745
3. Almansouri, A. Y., Li, J., **Bumbulis, L.**, Alhadari, A., Colaco, K., Harvey, P., Akhtari, S., Chandran, V., Gladman, D. D., Cook, R. J., & Eder, L. (2026). Incidence rate and risk factors of arrhythmias in patients with psoriatic arthritis. *RMD Open*, *12* (1), e005512. DOI:10.1136/rmdopen-2025-005512
2. Colaco, K., **Bumbulis, L.**, Chandran, V., Cook, R., Gladman, D., & Eder, L. (2025). The effect of biologic therapies on serum metabolic biomarkers in patients with psoriatic arthritis. *The Journal of Rheumatology*, *52* (Suppl 2), 27–28. DOI:10.3899/jrheum.2025-0314.TOUR2C
1. **Bumbulis, L. S.**, & Cook, R. J. (2025). Robustness and efficiency considerations when testing process reliability with a limit of detection. *Mathematics*, *13* (8), 1274. DOI:10.3390/math13081274

External Research Funding

NSERC Canada Graduate Scholarship – Doctoral	2025–2028	$ 120,000
Ontario Graduate Scholarship (OGS) / QEII-GSST (Declined)	2025–2026	$ 15,000

Scholarships & Awards

Conference Presentation Awards

SSC Student Research Presentation Award: Biostatistics Competition	2025	$ 500
WSSAF Presentation Award	2024	$ 500

University of Waterloo Scholarships & Awards

Doctoral

President's Graduate Scholarship	2025–2028	$ 30,000

A list of acronyms can be found on page 4.

Ontario Graduate Fellowship	2026	$ 4,000
University of Waterloo Graduate Scholarship (×2)	2025–2026	$ 2,000
SAS Chair's Graduate Award (×2)	2025–2026	$ 1,500
SAS Department Comprehensive Examination Scholarship	2025	$ 1,000
Russell Hiscock Graduate Scholarship in SAS	2025	$ 2,000
SAS Department Teaching Assistant Award	2024	$ 500
Cotton Family Women in Mathematics Graduate Scholarship	2024	$ 3,000
Provost Doctoral Entrance Award for Women	2024	$ 5,000
Masters		
SAS Department Outstanding Academic Performance Award	2024	$ 500
University of Waterloo Graduate Scholarship (×2)	2024	$ 3,000
D.A. Sprott Entrance Award	2023	$ 1,000
Undergraduate		
President's International Experience Award	2022	$ 1,500
Faculty of Mathematics Scholarship	2017–2022	$ 10,000
Co-op Student of the Year, Faculty of Mathematics	2021	
U Sports Academic All-Canadian	2018–2021	
President's Research Award	2018	$ 1,500
President's Scholarship of Distinction	2017	$ 2,000

Presentations

Contributed Conference Presentations

6. *Multistate Joint Modelling of Marker and Failure Data under Dual Censoring Schemes.* ISCB Annual Conference, Freiburg (Germany). September 2026.
5. *Challenges and Opportunities in the Analysis of Cervical Screening Data.* SSC Annual Meeting, Hamilton, ON. June 2026.
4. *Joint Models for Marker and Failure Data under Dual Censoring Schemes.* ENAR Spring Meeting, Indianapolis, IN (USA). March 2026.
3. *Dual Censoring Schemes for Marker and Failure Time Processes.* SSC Annual Meeting, Saskatoon, SK. May 2025.
2. *Robustness and Efficiency Considerations when Testing Process Reliability with a Limit of Detection.* WSSAF, Waterloo, ON. October 2024.
1. *Testing Process Reliability under a Limit of Detection: Issues of Robustness and Efficiency.* SSC Annual Meeting, St. John's, NL. June 2024.

Teaching Experience

Guest Lectures & Exam Reviews

STAT 431: Generalized Linear Models	Spring 2026
Guest lecture on overdispersion (80 mins)	
Instructor: Cecilia A. Cotton	
STAT 450: Estimation and Hypothesis Testing	Winter 2025
Guest lecture on Bayesian estimation (50 mins)	
Instructor: Peter W. MacDonald	
STAT 850: Estimation and Hypothesis Testing (Graduate-Level)	Fall 2024
Two final exam review sessions (3 hours each)	
Instructor: Yingli Qin	

Teaching Assistantships

STAT 231: Statistics	Spring 2025
STAT 230: Probability (Online)	Winter 2025
STAT 450: Estimation and Hypothesis Testing	Winter 2025
STAT 330: Mathematical Statistics †	Fall 2024

Credentials

Foundations for University Teaching in SAS Certificate — Winter 2025

† *Received SAS Department TA award.*

All teaching experience affiliated with the University of Waterloo; see course descriptions here.

Professional Activities

External Roles

Member of CANSSI Ontario Student PhD and Research Network (SPARK) — 2025–2026

Organizer of 2026 CANSSI Ontario STatistics (CAST) seminar by Kuan Liu

Internal Roles

Co-organizer for SAS Student Seminar Series — 2025–present

Graduate representative for SAS Department (elected) — 2025–2027

Co-Chair of 6th Waterloo Student Conference in SAS & Finance — 2026

Graduate representative for Women in Mathematics Committee — 2024–2025

Co-organizer for 2025 Navigating Your Careers Conference, University of Waterloo

Co-organizer for 2025 Ontario Graduate Mathematics Conference, University of Toronto

Conference Session Chair

Invited Sessions

SSC Annual Meeting: *Design & Analysis of Two-Phase Studies* — 2027

Contributed Sessions

ENAR Spring Meeting: *Artificial Intelligence for Biomedical Decision-Making* — 2026

WSSAF: *Methods for Inference and Improving Efficiency* — 2024

Work Experience

Data Analyst / Software Developer (Full-time), Emmetros — Jan.–Apr. 2023

Data Scientist (Co-op), Omnium — Jan.–Apr. 2021, May–Aug. 2022

Nominated for 2022 CEIA Student Achievement Award

Bioinformatics Analyst (Co-op), Ontario Institute for Cancer Research — May–Aug. 2021

Awarded 2021 Co-op Student of the Year, Faculty of Mathematics

Data Analyst / Software Developer (Co-op/Part-time), Emmetros — Sep. 2019–Dec. 2020

Nominated for 2020 Co-op Student of the Year, Faculty of Mathematics

Additional Activities

Intramural Sports, University of Waterloo

7v7 Soccer — Spring 2026

Volleyball (Team Captain) — Winter 2025

Flag Football — Spring 2024

Varsity Swim Team, University of Waterloo

Volunteer Assistant Coach — 2021–2022

Co-Captain — 2020–2021

Member — 2017–2021

Music

Piano Player — 2007–present

RCM Level 9 Harmony Certificate — 2013

RCM Level 7 Piano Certificate — 2013

Acronyms

ENAR	Eastern North American Region (International Biometric Society)
CANSSI	Canadian Statistical Sciences Institute
CEIA	Canadian Education & Internship Association
ISCB	International Society for Clinical Biostatistics
NSERC	Natural Sciences & Engineering Research Council
QEII-GSST	Queen Elizabeth II Graduate Scholarship in Science & Technology
RCM	Royal Conservatory of Music
RMD	Rheumatic & Musculoskeletal Diseases
SAS	Statistics & Actuarial Science
SSC	Statistical Society of Canada
WSSAF	Waterloo Student Conference in Statistics, Actuarial Science, & Finance